ACADEMIC CURRICULUM VITAE

SK Rakib Ul Islam Rahat

PhD Student in Computer Science | Wright State University

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Portfolio | Google Scholar | GitHub | ORCID 0009-0005-0744-8398

RESEARCH FOCUS

Trustworthy AI at the intersection of medical imaging, reliability under dataset shift, calibration, semantic transport, knowledge engineering, and knowledge-guided/neurosymbolic reasoning.

Doctoral advisor: Dr. Cogan Shimizu

EDUCATION

PhD, Computer Science

Wright State University · 2026-Present

MBA, Management Information Systems

International American University · 2024-2026

MSc, Global Business and Administration

Kyungsung University · 2022-2023

B.Tech, Computer Science and Engineering

Lovely Professional University · 2017-2021

RESEARCH & PROFESSIONAL EXPERIENCE

PhD Student / Researcher

Wright State University · 2026-Present · Dayton, Ohio

Doctoral research connecting trustworthy medical AI, reliable evaluation, knowledge engineering, and semantic representations.

Independent Medical AI Researcher

2023-Present · United States

Developed medical-imaging studies on dataset shift, calibration, shortcut learning, and reproducible external evaluation.

AI & Data Intern

Taskimpetus · 2024-2026

Supported analytics and automation using Python and SQL, exploratory analysis, and reproducible data workflows.

Teaching Assistant

Kyungsung University · 2022-2023

Mentored students on preprocessing, evaluation, and reproducible ML workflows; contributed to co-authored research.

Junior Software Engineer

KCIT · 2020-2021

Contributed to software engineering, smart-system integration, and applied technical development.

METHODS & TOOLS

Programming: Python, SQL, Java

Machine learning: PyTorch, TensorFlow, scikit-learn, CNNs, multimodal learning

Evaluation: External validation, calibration, dataset shift, Grad-CAM, SHAP

Research tools: Jupyter, Google Colab, GitHub, LaTeX, Overleaf

Current doctoral interests: Knowledge engineering, semantic representations, knowledge graphs, neurosymbolic AI

SK Rakib Ul Islam Rahat | Published research

JOURNAL ARTICLES

1. **Multi-stream contextual and frequency-aware deep learning with evidential uncertainty for Alzheimer's MRI slice classification**
Discover Artificial Intelligence (Springer Nature), 6, Article 1375 | 2026
https://doi.org/10.1007/s44163-026-02337-2

2. **TriGWONet: A lightweight multibranch convolutional neural network using gray wolf optimization for accurate oral cancer image classification**
Discover Artificial Intelligence (Springer Nature) | 2026
https://doi.org/10.1007/s44163-025-00776-x

3. **Multimodal Deep Learning for Classifying Diabetic Retinopathy Severity**
Journal of Korea Multimedia Society | 2026
https://doi.org/10.9717/kmms.2026.29.3.524

4. **Advancing diabetic retinopathy detection with AI and deep learning: Opportunities, limitations, and clinical barriers**
British Journal of Nursing Studies | 2025
https://doi.org/10.32996/bjns.2025.5.2.1

5. **Artificial intelligence for chronic kidney disease risk stratification in the USA: Ensemble vs. deep learning methods**
British Journal of Nursing Studies | 2025
https://doi.org/10.32996/bjns.2025.5.2.3

6. **A Deep Learning Framework for Early Breast Cancer Detection Among US Women: Integrating Mammography and Clinical EHR Data**
British Journal of Nursing Studies | 2025
https://doi.org/10.32996/bjns.2025.5.2.6

7. **Machine Learning-Based Hospital Readmission Prediction and Risk Analysis in the United States Healthcare System**
Journal of Computer Science and Technology Studies (JCSTS) | 2024
https://doi.org/10.32996/jcsts.2024.6.5.32

8. **Deep Learning-Based Skin Cancer Diagnosis in the United States: Advances, Challenges, and Clinical Translation**
Journal of Medical and Health Studies | 2023
https://doi.org/10.32996/jmhs.2023.4.6.18

9. **A Transfer Learning-Based Deep Convolutional Neural Network Framework for Automated Multi-Class Eye Disease Classification in the USA Using Retinal Fundus Image**
Journal of Medical and Health Studies | 2023
https://doi.org/10.32996/jmhs.2023.4.4.24

CONFERENCE PAPERS

1. **Performance Evaluation of Hybrid Machine Learning Models for Heart Disease Prediction in U.S. Clinical Decision Support Systems**
IEEE International Conference on Sentiment Analysis and Deep Learning (ICSADL) | 2026
https://doi.org/10.1109/ICSADL67539.2026.11452047

2. **An ANN Network-Based Approach for Early Detection of Parkinson's Disease Through Image Processing**
IEEE Conference on Converging Technology | 2025
https://doi.org/10.1109/ICCTEIE66144.2025.11341843

3. **PneuNet: A Multi-Scale Attention-Enhanced CNN for Pediatric Pneumonia Detection from Chest X-rays**
IEEE DELCON | 2025
https://doi.org/10.1109/DELCON68055.2025.11400229

4. **Deep Learning-Based Multi-Class Brain Tumor Classification from MRI using a Novel MS-DSCCNet Architecture**
IEEE DELCON | 2025
https://doi.org/10.1109/DELCON68055.2025.11400450

SK Rakib Ul Islam Rahat | Research & academic service

ACCEPTED FOR PUBLICATION

1. **Calibration Failure and Limited-Label Recalibration of Retinal Image Classifiers Under Dataset Shift**
Discover Artificial Intelligence (Springer Nature) | Accepted October 8, 2026

MANUSCRIPTS

1. **Reduced background sensitivity does not ensure external discrimination in fundus classifiers**
Biomedical Signal Processing and Control | Submitted | 2026
Submitted September 2026.

2. **Evaluation of lightweight AI models for early diabetic retinopathy detection in resource-constrained screening settings**
In review | Under Review | 2026
Manuscript under review.

SELECTED RESEARCH PROJECTS

Background sensitivity and external discrimination in fundus classifiers

Twelve models. Two architectures. One fixed APTOS split. A study separating background sensitivity from external discrimination on a processed Messidor-2 archive.

Exposing_Dataset_Artifacts

Semantic transport across chest X-ray benchmarks

Five-seed DenseNet-121 experiments across Kaggle, RSNA, and CheXpert examine label transport, calibration, and high-confidence errors.

semantic-shift-chest-xray

Calibration-aware diabetic-retinopathy reliability

Traceable experiments on calibration transfer from APTOS to Messidor-2, preserving split definitions, predictions, metrics, and audit artifacts.

calibration-aware-dr-reliability

Lightweight diabetic-retinopathy screening models

Five-fold comparison of EfficientNet-B0, MobileNetV2, and SqueezeNet using out-of-fold predictions and paired statistical tests.

Lightweight_DR_Detection_Models

Provenance-aware analytics across chest radiograph tasks

Compares discrimination, calibration, high-confidence errors, and source-threshold flag volume across Kaggle/Kermany pneumonia, RSNA lung opacity, and CheXpert report-derived tasks. The study treats microbiology as a future workflow boundary; it does not claim pathogen prediction.

chest-xray-microbiology-analytics

PEER REVIEW & SERVICE

10 completed reviews · 5 journals

Includes completed review rounds. Updated September 17, 2026.

- IAES International Journal of Robotics and Automation (IJRA)
- International Journal of Electrical and Computer Engineering (IJECE)
- International Journal of Power Electronics and Drive Systems (IJPEDS)
- Computer Methods in Biomechanics and Biomedical Engineering
- Information Fusion

SELECTED CERTIFICATIONS

- IBM Data Science
- Google Cloud AI Healthcare
- Illinois Tech AI Healthcare
- Elsevier Researcher Academy - Certified Peer Reviewer Course